

SDN Practical 7: Implementing BGP Communities

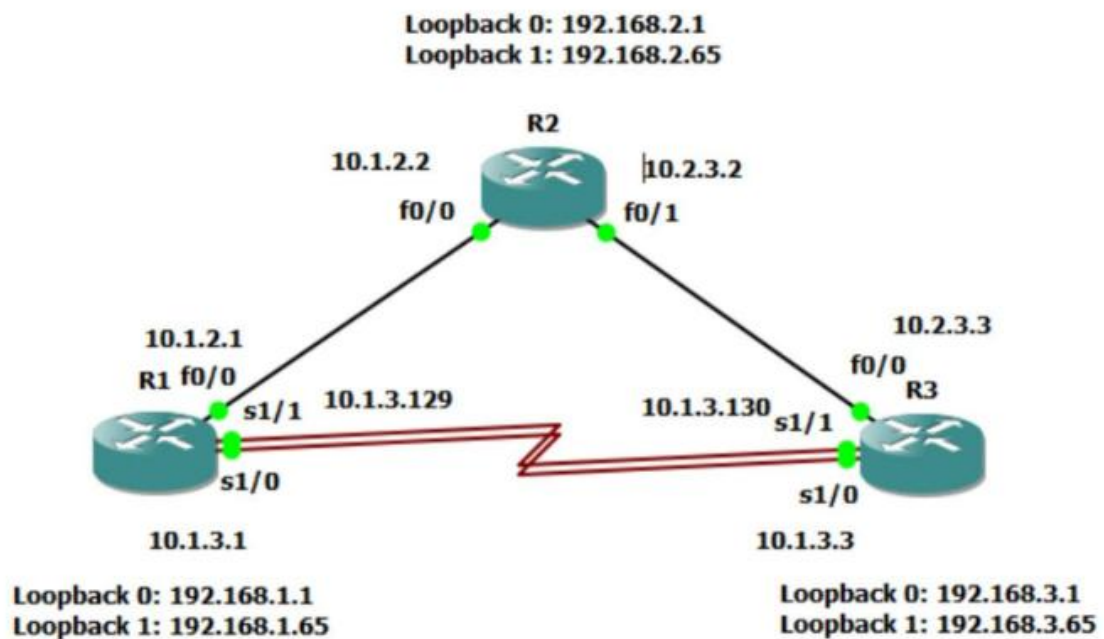
Aim: Implement BGP Communities

1. Implement eBGP for Ipv4.
2. Implement MP (Multi protocol) -BGP
3. Implement BGP path Manipulation

1. Implement eBGP for Ipv4.

Part 1: Build the Network and Configure Basic Device Settings and Interface Addressing

Step 1: Design the Topology



Step 2: Configure all 3 Routers.

• Router R1

no ip domain lookup

line console 0

logging synchronous

exec-timeout 0 0

exit

```
interface Loopback0
```

```
ip address 192.168.1.1 255.255.255.224
```

```
no shutdown
```

```
exit
```

```
interface Loopback1
```

```
ip address 192.168.1.65 255.255.255.192
```

```
no shutdown
```

```
exit
```

```
interface FastEthernet0/0
```

```
ip address 10.1.2.1 255.255.255.0
```

```
speed auto
```

```
duplex auto
```

```
no shutdown
```

```
exit
```

```
interface Serial1/0
```

```
ip address 10.1.3.1 255.255.255.128
```

```
clock rate 64000
```

```
encapsulation ppp
```

```
no shutdown
```

```
exit
```

interface Serial1/1

ip address 10.1.3.129 255.255.255.128

clock rate 64000

encapsulation ppp

no shutdown

exit

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#no ip domain lookup
R1(config)#
R1(config)#line console 0
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 0 0
R1(config-line)#exit
R1(config)#
R1(config)#interface Loopback0
R1(config-if)#ip address 192.168.1.1 255.255.255.224
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#interface Loopback1
R1(config-if)#ip address 192.168.1.65 255.255.255.192
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 10.1.2.1 255.255.255.0
R1(config-if)#speed auto
R1(config-if)#
^
% Invalid input detected at '^' marker.

R1(config-if)#duplex auto
R1(config-if)#
^
% Invalid input detected at '^' marker.

R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#interface Serial1/0
R1(config-if)#ip address 10.1.3.1 255.255.255.128
R1(config-if)#clock rate 64000
R1(config-if)#encapsulation ppp
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#interface Serial1/1
R1(config-if)#ip address 10.1.3.129 255.255.255.128
R1(config-if)#clock rate 64000
R1(config-if)#encapsulation ppp
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
```

Router 2:

no ip domain lookup

line console 0
logging synchronous
exec-timeout 0 0
exit

interface Loopback0
ip address 192.168.2.1 255.255.255.224
no shutdown
exit

interface Loopback1
ip address 192.168.2.65 255.255.255.192
no shutdown
exit

interface FastEthernet0/0
ip address 10.1.2.2 255.255.255.0
speed auto
duplex auto
no shutdown
exit

interface FastEthernet0/1
ip address 10.2.3.2 255.255.255.0
speed auto
duplex auto
no shutdown
exit

```

R2#
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#no ip domain lookup
R2(config)#
R2(config)#line console 0
R2(config-line)#logging synchronous
R2(config-line)#exec-timeout 0 0
R2(config-line)#exit
R2(config)#
R2(config)#interface Loopback0
R2(config-if)#ip address 192.168.2.1 255.255.255.224
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
R2(config)#interface Loopback1
R2(config-if)#ip address 192.168.2.65 255.255.255.192
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
R2(config)#interface FastEthernet0/0
R2(config-if)#ip address 10.1.2.2 255.255.255.0
R2(config-if)#speed auto
R2(config-if)#duplex auto
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
R2(config)#interface FastEthernet0/1
R2(config-if)#ip address 10.2.3.2 255.255.255.0
R2(config-if)#speed auto
R2(config-if)#duplex auto
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Nov 11 14:39:45.631: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0,
  changed state to up
*Nov 11 14:39:45.711: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1,
  changed state to up
R2(config)#
*Nov 11 14:39:47.803: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Nov 11 14:39:47.851: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
o up
*Nov 11 14:39:48.803: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R2(config)#
*Nov 11 14:39:48.851: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/1, changed state to up
R2(config)#

```

Router 3:

```

conf t
no ip domain lookup

line console 0
logging synchronous
exec-timeout 0 0
exit

```

```
interface Loopback0
ip address 192.168.3.1 255.255.255.224
no shutdown
exit
```

```
interface Loopback1
ip address 192.168.3.65 255.255.255.192
no shutdown
exit
```

```
interface FastEthernet0/0
ip address 10.2.3.3 255.255.255.0
speed auto
duplex auto
no shutdown
exit
```

```
interface Serial1/0
ip address 10.1.3.3 255.255.255.128
clock rate 64000
encapsulation ppp
no shutdown
exit
```

```
interface Serial1/1
ip address 10.1.3.131 255.255.255.128
clock rate 64000
encapsulation ppp
no shutdown
exit
```

```

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#no ip domain lookup
R3(config)#
R3(config)#line console 0
R3(config-line)#logging synchronous
R3(config-line)#exec-timeout 0 0
R3(config-line)#exit
R3(config)#
R3(config)#interface Loopback0
R3(config-if)#ip address 192.168.3.1 255.255.255.224
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
R3(config)#interface Loopback1
R3(config-if)#ip address 192.168.3.65 255.255.255.192
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
R3(config)#interface FastEthernet0/0
R3(config-if)#ip address 10.2.3.3 255.255.255.0
R3(config-if)#speed auto
R3(config-if)#
^
% Invalid input detected at '^' marker.

R3(config-if)#duplex auto
R3(config-if)#
^
% Invalid input detected at '^' marker.

R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
R3(config)#interface Serial1/0
R3(config-if)#ip address 10.1.3.3 255.255.255.128
R3(config-if)#clock rate 64000
R3(config-if)#encapsulation ppp
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
R3(config)#interface Serial1/1
R3(config-if)#ip address 10.1.3.131 255.255.255.128
R3(config-if)#clock rate 64000
R3(config-if)#encapsulation ppp
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Nov 11 14:43:21.163: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0,
changed state to up
*Nov 11 14:43:21.239: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1,
changed state to up
*Nov 11 14:43:22.151: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
*Nov 11 14:43:22.167: %LINK-3-UPDOWN: Interface Serial1/1, changed state to up
R3(config)#
*Nov 11 14:43:22.183: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
*Nov 11 14:43:22.231: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to up
R3(config)#
*Nov 11 14:43:23.347: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Nov 11 14:43:24.347: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R3(config)#

```

Part 2: Configure and Verify eBGP for IPv4 on all Routers

Step 1: Implement BGP and neighbor relationships on R1.

Router 1:

router bgp 1000

bgp router-id 1.1.1.1

neighbor 10.1.2.2 remote-as 500

neighbor 10.1.3.3 remote-as 300

network 192.168.1.0 mask 255.255.255.224

network 192.168.1.64 mask 255.255.255.192

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router bgp 1000
R1(config-router)#bgp router-id 1.1.1.1
R1(config-router)#neighbor 10.1.2.2 remote-as 500
R1(config-router)#neighbor 10.1.3.3 remote-as 300
R1(config-router)#network 192.168.1.0 mask 255.255.255.224
R1(config-router)#network 192.168.1.64 mask 255.255.255.192
R1(config-router)#
```

Step 2: Implement BGP and neighbor relationships on R2.

R2:

```
router bgp 500
bgp router-id 2.2.2.2
neighbor 10.1.2.1 remote-as 1000
neighbor 10.2.3.3 remote-as 300
network 192.168.2.0 mask 255.255.255.224
network 192.168.2.64 mask 255.255.255.192
```

```
R2(config)#router bgp 500
R2(config-router)#bgp router-id 2.2.2.2
R2(config-router)#neighbor 10.1.2.1 remote-as 1000
R2(config-router)#neighbor 10.2.3.3 remote-as 300
R2(config-router)#network 192.168.2.0 mask 255.255.255.224
R2(config-router)#network 192.168.2.64 mask 255.255.255.192
R2(config-router)#
R2(config-router)#
*Nov 11 14:49:14.739: %BGP-5-ADJCHANGE: neighbor 10.1.2.1 Up
R2(config-router)#
```

Step 3: Implement BGP and neighbor relationships on R3

R3:

```
router bgp 300
bgp router-id 3.3.3.3
no bgp default ipv4-unicast
neighbor 10.2.3.2 remote-as 500
neighbor 10.1.3.1 remote-as 1000
neighbor 10.1.3.129 remote-as 1000
```

```
R3(config)#router bgp 300
R3(config-router)#bgp router-id 3.3.3.3
R3(config-router)#no bgp default ipv4-unicast
R3(config-router)#neighbor 10.2.3.2 remote-as 500
R3(config-router)#neighbor 10.1.3.1 remote-as 1000
R3(config-router)#neighbor 10.1.3.129 remote-as 1000
R3(config-router)#
```

Step 4: Verifying BGP neighbor relationships.

```
R1#show ip route bgp
```



```

R1#show ip route bgp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
B       192.168.2.0/27 [20/0] via 10.1.2.2, 00:03:22
B       192.168.2.64/26 [20/0] via 10.1.2.2, 00:03:22
R1#
*Nov 11 14:53:46.339: %SYS-5-CONFIG_I: Configured from console by console

```

R2#show ip route bgp

```

R2#show ip route bgp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
B       192.168.1.0/27 [20/0] via 10.1.2.1, 00:04:00
B       192.168.1.64/26 [20/0] via 10.1.2.1, 00:04:00
R2#
*Nov 11 14:54:24.231: %SYS-5-CONFIG_I: Configured from console by console
R2#

```

R2#show ip bgp neighbors

```

R2#show ip bgp neighbors
BGP neighbor is 10.1.2.1, remote AS 1000, external link
  BGP version 4, remote router ID 1.1.1.1
  BGP state = Established, up for 00:05:44
  Last read 00:00:50, last write 00:00:57, hold time is 180, keepalive interval is 60 seconds
  Neighbor sessions:
    1 active, is not multisession capable (disabled)
  Neighbor capabilities:
    Route refresh: advertised and received(new)
    Four-octets ASN Capability: advertised and received
    Address family IPv4 Unicast: advertised and received
    Enhanced Refresh Capability: advertised and received
    Multisession Capability:
    Stateful switchover support enabled: NO for session 1
  Message statistics:
    InQ depth is 0
    OutQ depth is 0

      Sent      Rcvd
Opens:          1          1
Notifications:  0          0
Updates:        2          2
Keepalives:     6          6
Route Refresh:  0          0
Total:          9          9
Default minimum time between advertisement runs is 30 seconds

For address family: IPv4 Unicast
Session: 10.1.2.1
BGP table version 5, neighbor version 5/0
Output queue size : 0
Index 1, Advertise bit 0
1 update-group member
Slow-peer detection is disabled
Slow-peer split-update-group dynamic is disabled

      Sent      Rcvd
Prefix activity: ----
Prefixes Current:      2          2 (Consumes 160 bytes)
Prefixes Total:        2          2
Implicit Withdraw:      0          0
Explicit Withdraw:      0          0
Used as bestpath:      n/a          2
Used as multipath:      n/a          0

      Outbound   Inbound
Local Policy Denied Prefixes: -----
Bestpath from this peer:      2          n/a
Total:                        2          0
Number of NLRI in the update sent: max 2, min 0
Last detected as dynamic slow peer: never
Dynamic slow peer recovered: never
Refresh Epoch: 1
Last Sent Refresh Start-of-rib: never
Last Sent Refresh End-of-rib: never
Last Received Refresh Start-of-rib: never
Last Received Refresh End-of-rib: never

      Sent      Rcvd
Refresh activity: ----
Refresh Start-of-RIB      0          0
Refresh End-of-RIB        0          0
--More--

```

The interfaces on R3 need to be activated in IPv4 AF configuration mode

Router 3:

router bgp 300

bgp router-id 3.3.3.3

no bgp default ipv4-unicast

```
neighbor 10.2.3.2 remote-as 500
neighbor 10.1.3.1 remote-as 1000
neighbor 10.1.3.129 remote-as 1000
```

```
address-family ipv4
  neighbor 10.1.3.1 activate
  neighbor 10.1.3.129 activate
  neighbor 10.2.3.2 activate
  network 192.168.3.0 mask 255.255.255.224
  network 192.168.3.64 mask 255.255.255.192
exit-address-family
```

```
R3(config-router)#address-family ipv4
R3(config-router-af)# neighbor 10.1.3.1 activate
R3(config-router-af)# neighbor 10.1.3.129 activate
R3(config-router-af)# neighbor 10.2.3.2 activate
R3(config-router-af)# network 192.168.3.0 mask 255.255.255.224
R3(config-router-af)# network 192.168.3.64 mask 255.255.255.192
R3(config-router-af)#
*Nov 11 14:57:44.959: %BGP-5-ADJCHANGE: neighbor 10.2.3.2 Up
R3(config-router-af)#
R3(config-router-af)#
*Nov 11 14:57:47.207: %BGP-5-ADJCHANGE: neighbor 10.1.3.1 Up
R3(config-router-af)#
```

Verify that the BGP state between R2 and R3 has now been established

R2#show ip bgp neighbors | begin BGP neighbor is 10.2.3.3

```
R2#show ip bgp neighbors | begin BGP neighbor is 10.2.3.3
BGP neighbor is 10.2.3.3, remote AS 300, external link
  BGP version 4, remote router ID 3.3.3.3
  BGP state = Established, up for 00:01:59
  Last read 00:00:40, last write 00:00:13, hold time is 180, keepalive interval is 60 seconds
  Neighbor sessions:
    1 active, is not multisession capable (disabled)
  Neighbor capabilities:
    Route refresh: advertised and received(new)
    Four-octets ASN Capability: advertised and received
    Address family IPv4 Unicast: advertised and received
    Enhanced Refresh Capability: advertised and received
    Multisession Capability:
    Stateful switchover support enabled: NO for session 1
  Message statistics:
    InQ depth is 0
    OutQ depth is 0

              Sent          Rcvd
  Opens:              1            1
  Notifications:      0            0
  Updates:             4            4
  Keepalives:          4            3
  Route Refresh:       0            0
  Total:              11            8
  Default minimum time between advertisement runs is 30 seconds
```

Step 5: Examining the running-configs.

R1#show running-config | section bgp

```
R1#show running-config | section bgp
router bgp 1000
  bgp router-id 1.1.1.1
  bgp log-neighbor-changes
  network 192.168.1.0 mask 255.255.255.224
  network 192.168.1.64 mask 255.255.255.192
  neighbor 10.1.2.2 remote-as 500
  neighbor 10.1.3.3 remote-as 300
R1#
```

```
R2#show running-config | section bgp
router bgp 500
  bgp router-id 2.2.2.2
  bgp log-neighbor-changes
  network 192.168.2.0 mask 255.255.255.224
  network 192.168.2.64 mask 255.255.255.192
  neighbor 10.1.2.1 remote-as 1000
  neighbor 10.2.3.3 remote-as 300
R2#
```

```
R3#show running-config | section bgp
router bgp 300
  bgp router-id 3.3.3.3
  bgp log-neighbor-changes
  no bgp default ipv4-unicast
  neighbor 10.1.3.1 remote-as 1000
  neighbor 10.1.3.129 remote-as 1000
  neighbor 10.2.3.2 remote-as 500
  !
  address-family ipv4
    network 192.168.3.0 mask 255.255.255.224
    network 192.168.3.64 mask 255.255.255.192
    neighbor 10.1.3.1 activate
    neighbor 10.1.3.129 activate
    neighbor 10.2.3.2 activate
  exit-address-family
R3#
```

Step 6: Verifying BGP operations.

```
R2#sh ip bgp
BGP table version is 7, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop           Metric LocPrf Weight Path
*   192.168.1.0/27    10.2.3.3                0 300 1000 i
*>                10.1.2.1                0 1000 i
*   192.168.1.64/26   10.2.3.3                0 300 1000 i
*>                10.1.2.1                0 1000 i
*>   192.168.2.0/27    0.0.0.0                0 32768 i
*>   192.168.2.64/26   0.0.0.0                0 32768 i
*   192.168.3.0/27    10.1.2.1                0 1000 300 i
*>                10.2.3.3                0 300 i
*   192.168.3.64/26   10.1.2.1                0 1000 300 i
*>                10.2.3.3                0 300 i
```

sh ip bgp 192.168.1.0

```
R2#sh ip bgp 192.168.1.0
BGP routing table entry for 192.168.1.0/27, version 4
Paths: (2 available, best #2, table default)
  Advertised to update-groups:
    1
  Refresh Epoch 1
  300 1000
    10.2.3.3 from 10.2.3.3 (3.3.3.3)
      Origin IGP, localpref 100, valid, external
      rx pathid: 0, tx pathid: 0
  Refresh Epoch 1
  1000
    10.1.2.1 from 10.1.2.1 (1.1.1.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
```

```
R2#sh ip bgp 192.168.1.0
BGP routing table entry for 192.168.1.0/27, version 4
Paths: (2 available, best #2, table default)
  Advertised to update-groups:
    1
  Refresh Epoch 1
  300 1000
    10.2.3.3 from 10.2.3.3 (3.3.3.3)
      Origin IGP, localpref 100, valid, external
      rx pathid: 0, tx pathid: 0
  Refresh Epoch 1
  1000
    10.1.2.1 from 10.1.2.1 (1.1.1.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
R2#show ip bgp neighbors
BGP neighbor is 10.1.2.1, remote AS 1000, external link
  BGP version 4, remote router ID 1.1.1.1
  BGP state = Established, up for 00:17:27
  Last read 00:00:28, last write 00:00:48, hold time is 180, keepalive interval is 60 seconds
  Neighbor sessions:
    1 active, is not multisession capable (disabled)
  Neighbor capabilities:
    Route refresh: advertised and received(new)
    Four-octets ASN Capability: advertised and received
    Address family IPv4 Unicast: advertised and received
    Enhanced Refresh Capability: advertised and received
    Multisession Capability:
    Stateful switchover support enabled: NO for session 1
  Message statistics:
    InQ depth is 0
    OutQ depth is 0

              Sent          Rcvd
Opens:          1             1
Notifications:  0             0
Updates:         3             3
Keepalives:     18            19
Route Refresh:   0             0
Total:          22            23
Default minimum time between advertisement runs is 30 seconds

For address family: IPv4 Unicast
Session: 10.1.2.1
BGP table version 7, neighbor version 7/0
Output queue size : 0
Index 1, Advertise bit 0
1 update-group member
Slow-peer detection is disabled
Slow-peer split-update-group dynamic is disabled

              Sent          Rcvd
Prefix activity:
----
Prefixes Current:    6             4 (Consumes 320 bytes)
Prefixes Total:      8             4
Implicit Withdraw:    2             0
Explicit Withdraw:    0             0
Used as bestpath:    n/a             2
Used as multipath:    n/a             0

Outbound  Inbound
--More--
```

show ip bgp neighbors

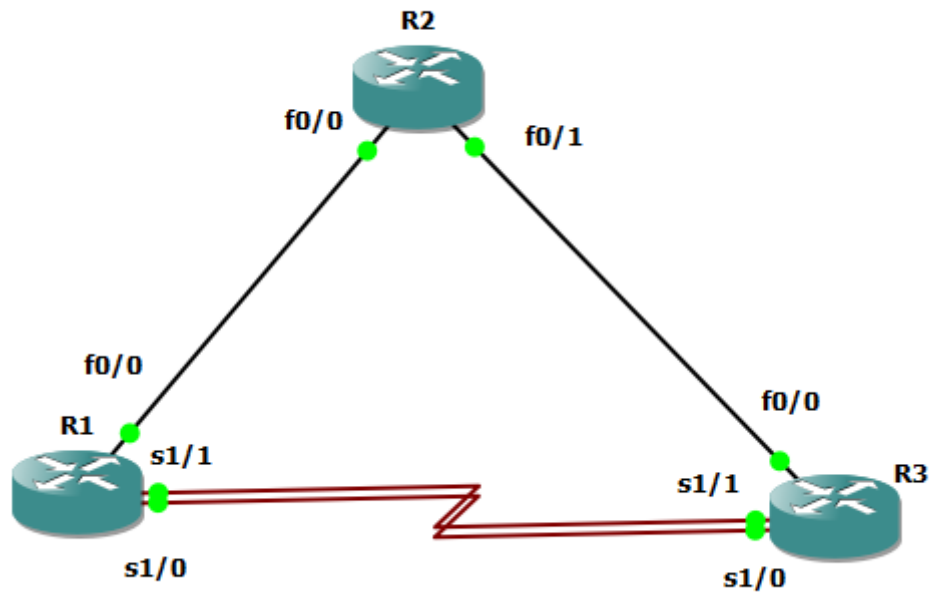
```
R2#show ip bgp neighbors
BGP neighbor is 10.1.2.1, remote AS 1000, external link
  BGP version 4, remote router ID 1.1.1.1
  BGP state = Established, up for 00:18:34
  Last read 00:00:40, last write 00:00:03, hold time is 180, keepalive interval is 60 seconds
  Neighbor sessions:
    1 active, is not multisession capable (disabled)
  Neighbor capabilities:
    Route refresh: advertised and received(new)
    Four-octets ASN Capability: advertised and received
    Address family IPv4 Unicast: advertised and received
    Enhanced Refresh Capability: advertised and received
    Multisession Capability:
    Stateful switchover support enabled: NO for session 1
  Message statistics:
    InQ depth is 0
    OutQ depth is 0

      Sent      Rcvd
  Opens:          1        1
  Notifications:    0        0
  Updates:         3        3
  Keepalives:      20       20
--More--
```

2. Implement MP(Multi protocol) -BGP

Part 1: Build the Network and Configure Basic Device Settings and Interface Addressing

Step 1:Design the Topology



Step 2: Configure basic settings for each router.

• **Router R1:**

```
conf t
no ip domain lookup
line con 0
exec-timeout 0 0
logging synchronous
exit
banner motd # This is R1, BGP Manipulation Lab #
ipv6 unicast-routing
```

```
interface f0/0
ip address 10.1.2.1 255.255.255.0
ipv6 address fe80::1:1 link-local
ipv6 address 2001:DB8:acad:1012::1/64
no shutdown
exit
```

```
interface s1/0
ip address 10.1.3.1 255.255.255.128
ipv6 address fe80::1:2 link-local
ipv6 address 2001:DB8:acad:1013::1/80
clock rate 64000
no keepalive
no shutdown
exit
```

```
interface s1/1
ip address 10.1.3.1 255.255.255.128
ipv6 address fe80::1:3 link-local
ipv6 address 2001:DB8:acad:1014::1/80
clock rate 64000
no keepalive
no shutdown
exit

end

wr
```

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#no ip domain lookup
R1(config)#line con 0
R1(config-line)# exec-timeout 0 0
R1(config-line)# logging synchronous
R1(config-line)#exit
R1(config)#banner motd # This is R1, BGP Manipulation Lab #
R1(config)#ipv6 unicast-routing
R1(config)#
R1(config)#interface f0/0
R1(config-if)# ip address 10.1.2.1 255.255.255.0
R1(config-if)# ipv6 address fe80::1:1 link-local
R1(config-if)# ipv6 address 2001:DB8:acad:1012::1/64
R1(config-if)# no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#interface s1/0
R1(config-if)# ip address 10.1.3.1 255.255.255.128
R1(config-if)# ipv6 address fe80::1:2 link-local
R1(config-if)# ipv6 address 2001:DB8:acad:1013::1/80
R1(config-if)# clock rate 64000
R1(config-if)# no keepalive
R1(config-if)# no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#interface s1/1
R1(config-if)# ip address 10.1.3.1 255.255.255.128
R1(config-if)# ipv6 address fe80::1:3 link-local
R1(config-if)# ipv6 address 2001:DB8:acad:1014::1/80
R1(config-if)# clock rate 64000
R1(config-if)# no keepalive
R1(config-if)# no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#end
```

```
interface loopback0
ip address 192.168.1.1 255.255.255.224
ipv6 address fe80::1:4 link-local
ipv6 address 2001:DB8:acad:1000::1/64
no shutdown
exit
```



```
interface loopback1
ip address 192.168.1.65 255.255.255.192
ipv6 address fe80::1:5 link-local
ipv6 address 2001:DB8:acad:1001::1/64
no shutdown
exit

end

wr
```

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface loopback0
R1(config-if)# ip address 192.168.1.1 255.255.255.224
R1(config-if)# ipv6 address fe80::1:4 link-local
R1(config-if)# ipv6 address 2001:DB8:acad:1000::1/64
R1(config-if)# no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#interface loopback1
R1(config-if)# ip address 192.168.1.65 255.255.255.192
R1(config-if)# ipv6 address fe80::1:5 link-local
R1(config-if)# ipv6 address 2001:DB8:acad:1001::1/64
R1(config-if)# no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#end
```

Router 2:

```
conf t
no ip domain lookup
line con 0
exec-timeout 0 0
logging synchronous
exit
banner motd # This is R2, BGP Path Manipulation Lab #
ipv6 unicast-routing
```

```
interface f0/0
ip address 10.1.2.2 255.255.255.0
ipv6 address fe80::2:1 link-local
ipv6 address 2001:DB8:acad:1021::2/64
no shutdown
exit
```

```
interface f0/1
ip address 10.2.3.2 255.255.255.0
ipv6 address fe80::2:2 link-local
ipv6 address 2001:DB8:acad:1023::2/64
no shutdown
exit
```

```
interface loopback0
ip address 192.168.2.1 255.255.255.224
ipv6 address fe80::2:3 link-local
ipv6 address 2001:DB8:acad:2000::2/64
no shutdown
exit
```

```
interface loopback1
ip address 192.168.2.65 255.255.255.192
ipv6 address fe80::2:4 link-local
ipv6 address 2001:DB8:acad:2001::2/64
no shutdown
exit
```

```
end
wr
```

```
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#no ip domain lookup
R2(config)#line con 0
R2(config-line)# exec-timeout 0 0
R2(config-line)# logging synchronous
R2(config-line)#exit
R2(config)#banner motd # This is R2, BGP Path Manipulation Lab #
R2(config)#ipv6 unicast-routing
R2(config)#
R2(config)#interface f0/0
R2(config-if)# ip address 10.1.2.2 255.255.255.0
R2(config-if)# ipv6 address fe80::2:1 link-local
R2(config-if)# ipv6 address 2001:DB8:acad:1021::2/64
R2(config-if)# no shutdown
R2(config-if)#exit
R2(config)#
R2(config)#interface f0/1
R2(config-if)# ip address 10.2.3.2 255.255.255.0
R2(config-if)# ipv6 address fe80::2:2 link-local
R2(config-if)# ipv6 address 2001:DB8:acad:1023::2/64
R2(config-if)# no shutdown
R2(config-if)#exit
R2(config)#
R2(config)#interface loopback0
R2(config-if)# ip address 192.168.2.1 255.255.255.224
R2(config-if)# ipv6 address fe80::2:3 link-local
R2(config-if)# ipv6 address 2001:DB8:acad:2000::2/64
R2(config-if)# no shutdown
R2(config-if)#exit
R2(config)#
R2(config)#interface loopback1
R2(config-if)# ip address 192.168.2.65 255.255.255.192
R2(config-if)# ipv6 address fe80::2:4 link-local
R2(config-if)# ipv6 address 2001:DB8:acad:2001::2/64
R2(config-if)# no shutdown
R2(config-if)#exit
R2(config)#
R2(config)#end
R2#wr
```

Router R3:

```
conf t
no ip domain lookup
line con 0
exec-timeout 0 0
logging synchronous
```

```
exit
banner motd # This is R3, BGP Path Manipulation Lab #
ipv6 unicast-routing
```

```
interface f0/0
ip address 10.2.3.3 255.255.255.0
ipv6 address fe80::3:1 link-local
ipv6 address 2001:DB8:acad:1023::3/64
no shutdown
exit
```

```
interface s1/0
ip address 10.1.3.3 255.255.255.128
ipv6 address fe80::3:2 link-local
ipv6 address 2001:DB8:acad:1013::3/80
clock rate 64000
no keepalive
no shutdown
exit
```

```
interface s1/1
ip address 10.1.3.130 255.255.255.128
ipv6 address fe80::3:3 link-local
ipv6 address 2001:DB8:acad:1014::3/80
clock rate 64000
no keepalive
no shutdown
exit
```

```
end
wr
```

```
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#no ip domain lookup
R3(config)#line con 0
R3(config-line)# exec-timeout 0 0
R3(config-line)# logging synchronous
R3(config-line)#exit
R3(config)#banner motd # This is R3, BGP Path Manipulation Lab #
R3(config)#ipv6 unicast-routing
R3(config)#
R3(config)#interface f0/0
R3(config-if)# ip address 10.2.3.3 255.255.255.0
R3(config-if)# ipv6 address fe80::3:1 link-local
R3(config-if)# ipv6 address 2001:DB8:acad:1023::3/64
R3(config-if)# no shutdown
R3(config-if)#exit
R3(config)#
R3(config)#interface s1/0
R3(config-if)# ip address 10.1.3.3 255.255.255.128
R3(config-if)# ipv6 address fe80::3:2 link-local
R3(config-if)# ipv6 address 2001:DB8:acad:1013::3/80
R3(config-if)# clock rate 64000
R3(config-if)# no keepalive
R3(config-if)# no shutdown
R3(config-if)#exit
R3(config)#
R3(config)#interface s1/1
R3(config-if)# ip address 10.1.3.130 255.255.255.128
R3(config-if)# ipv6 address fe80::3:3 link-local
R3(config-if)# ipv6 address 2001:DB8:acad:1014::3/80
R3(config-if)# clock rate 64000
R3(config-if)# no keepalive
R3(config-if)# no shutdown
R3(config-if)#exit
R3(config)#
R3(config)#end
R3#wr
```

```
interface loopback0
ip address 192.168.3.1 255.255.255.224
ipv6 address fe80::3:4 link-local
ipv6 address 2001:DB8:acad:3000::1/64
no shutdown
exit
```

```
interface loopback1
ip address 192.168.3.65 255.255.255.192
ipv6 address fe80::3:5 link-local
ipv6 address 2001:DB8:acad:3001::1/64
no shutdown
exit
```

```

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#no ip domain lookup
R1(config)#line con 0
R1(config-line)# exec-timeout 0 0
R1(config-line)# logging synchronous
R1(config-line)#exit
R1(config)#banner motd # This is R1, BGP Manipulation Lab #
R1(config)#ipv6 unicast-routing
R1(config)#
R1(config)#interface f0/0
R1(config-if)# ip address 10.1.2.1 255.255.255.0
R1(config-if)# ipv6 address fe80::1:1 link-local
R1(config-if)# ipv6 address 2001:DB8:acad:1012::1/64
R1(config-if)# no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#interface s1/0
R1(config-if)# ip address 10.1.3.1 255.255.255.128
R1(config-if)# ipv6 address fe80::1:2 link-local
R1(config-if)# ipv6 address 2001:DB8:acad:1013::1/80
R1(config-if)# clock rate 64000
R1(config-if)# no keepalive
R1(config-if)# no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#interface s1/1
R1(config-if)# ip address 10.1.4.1 255.255.255.128
R1(config-if)# ipv6 address fe80::1:3 link-local
R1(config-if)# ipv6 address 2001:DB8:acad:1014::1/80
R1(config-if)# clock rate 64000
R1(config-if)# no keepalive
R1(config-if)# no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#interface loopback0
R1(config-if)# ip address 192.168.1.1 255.255.255.224
R1(config-if)# ipv6 address fe80::1:4 link-local
R1(config-if)# ipv6 address 2001:DB8:acad:1000::1/64
R1(config-if)# no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#interface loopback1
R1(config-if)# ip address 192.168.1.65 255.255.255.192
R1(config-if)# ipv6 address fe80::1:5 link-local
R1(config-if)# ipv6 address 2001:DB8:acad:1001::1/64
R1(config-if)# no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#end
R1#wr

```

Router R3:

```

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#interface loopback0
R3(config-if)# ip address 192.168.3.1 255.255.255.224
R3(config-if)# ipv6 address fe80::3:4 link-local
R3(config-if)# ipv6 address 2001:DB8:acad:3000::1/64
R3(config-if)# no shutdown
R3(config-if)#exit
R3(config)#
R3(config)#interface loopback1
R3(config-if)# ip address 192.168.3.65 255.255.255.192
R3(config-if)# ipv6 address fe80::3:5 link-local
R3(config-if)# ipv6 address 2001:DB8:acad:3001::1/64
R3(config-if)# no shutdown
R3(config-if)#exit
*Nov 12 15:18:39.535: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
*Nov 12 15:18:39.783: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1, changed state to up
R3(config-if)#exit
R3(config)#

```

Part 2: Configure and Verify Multi-Protocol BGP on all Routers

Step 1: On R1, create the core BGP configuration

conf t

router bgp 6500

bgp router-id 1.1.1.1

no bgp default ipv4-unicast

```
! IPv4 neighbors
neighbor 10.1.2.2 remote-as 500
neighbor 10.1.3.3 remote-as 300
neighbor 10.1.3.130 remote-as 300

! IPv6 neighbors
neighbor 2001:DB8:acad:1012::2 remote-as 500
neighbor 2001:DB8:acad:1013::3 remote-as 300
neighbor 2001:DB8:acad:1014::3 remote-as 300

end
wr
```

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router bgp 6500
R1(config-router)# bgp router-id 1.1.1.1
R1(config-router)# no bgp default ipv4-unicast
R1(config-router)#
R1(config-router)# ! IPv4 neighbors
R1(config-router)# neighbor 10.1.2.2 remote-as 500
R1(config-router)# neighbor 10.1.3.3 remote-as 300
R1(config-router)# neighbor 10.1.3.130 remote-as 300
R1(config-router)#
R1(config-router)# ! IPv6 neighbors
R1(config-router)# neighbor 2001:DB8:acad:1012::2 remote-as 500
R1(config-router)# neighbor 2001:DB8:acad:1013::3 remote-as 300
R1(config-router)# neighbor 2001:DB8:acad:1014::3 remote-as 300
R1(config-router)#
R1(config-router)#end
R1#wr
Building configuration...
[OK]
R1#
```

Step 2: On R1, configure the IPv4 unicast address family.

```
conf t
router bgp 6500
address-family ipv4 unicast
network 192.168.1.0 mask 255.255.255.224
network 192.168.1.64 mask 255.255.255.192
```

```
! Activate IPv4 BGP neighbors
neighbor 10.1.2.2 activate
neighbor 10.1.3.3 activate
neighbor 10.1.3.130 activate
exit
end
wr
```

```

R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router bgp 6500
R1(config-router)# address-family ipv4 unicast
R1(config-router-af)#  network 192.168.1.0 mask 255.255.255.224
R1(config-router-af)#  network 192.168.1.64 mask 255.255.255.192
R1(config-router-af)#
R1(config-router-af)#  ! Activate IPv4 BGP neighbors
R1(config-router-af)#  neighbor 10.1.2.2 activate
R1(config-router-af)#  neighbor 10.1.3.3 activate
R1(config-router-af)#  neighbor 10.1.3.130 activate
R1(config-router-af)# exit
R1(config-router)#end
R1#wr

```

Step 3: On R1, configure the IPv6 unicast address family.

```

conf t
router bgp 6500
address-family ipv6 unicast
network 2001:DB8:acad:1000::/64
network 2001:DB8:acad:1001::/64

! Activate IPv6 BGP neighbors
neighbor 2001:DB8:acad:1012::2 activate
neighbor 2001:DB8:acad:1013::3 activate
neighbor 2001:DB8:acad:1014::3 activate
exit
end
wr

```

```

R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router bgp 6500
R1(config-router)# address-family ipv6 unicast
R1(config-router-af)#  network 2001:DB8:acad:1000::/64
R1(config-router-af)#  network 2001:DB8:acad:1001::/64
R1(config-router-af)#
R1(config-router-af)#  ! Activate IPv6 BGP neighbors
R1(config-router-af)#  neighbor 2001:DB8:acad:1012::2 activate
R1(config-router-af)#  neighbor 2001:DB8:acad:1013::3 activate
R1(config-router-af)#  neighbor 2001:DB8:acad:1014::3 activate
R1(config-router-af)# exit
R1(config-router)#end
R1#wr

```

Step 4: Verify that MP-BGP is operational.

```

show bgp ipv4 unicast summary

```

```
R1#show bgp ipv4 unicast summary
BGP router identifier 1.1.1.1, local AS number 6500
BGP table version is 3, main routing table version 3
2 network entries using 288 bytes of memory
2 path entries using 160 bytes of memory
1/1 BGP path/bestpath attribute entries using 136 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 584 total bytes of memory
BGP activity 4/0 prefixes, 4/0 paths, scan interval 60 secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
10.1.2.2	4	500	0	0	1	0	0	never	Idle
10.1.3.3	4	300	0	0	1	0	0	never	Idle
10.1.3.130	4	300	0	0	1	0	0	never	Idle

R1#

```
R1#show bgp ipv6 unicast summary
BGP router identifier 1.1.1.1, local AS number 6500
BGP table version is 1, main routing table version 1
2 network entries using 336 bytes of memory
2 path entries using 208 bytes of memory
1/0 BGP path/bestpath attribute entries using 136 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 680 total bytes of memory
BGP activity 4/0 prefixes, 4/0 paths, scan interval 60 secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
2001:DB8:ACAD:1012::2	4	500	0	0	1	0	0	never	Active
2001:DB8:ACAD:1013::3	4	300	0	0	1	0	0	never	Idle
2001:DB8:ACAD:1014::3	4	300	0	0	1	0	0	never	Idle

R1#

```
R1#show bgp ipv4 unicast
BGP table version is 3, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found
```

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	192.168.1.0/27	0.0.0.0	0		32768	i
*>	192.168.1.64/26	0.0.0.0	0		32768	i

R1#

```
R1#show bgp ipv6 unicast
BGP table version is 1, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found
```

	Network	Next Hop	Metric	LocPrf	Weight	Path
*	2001:DB8:ACAD:1000::/64	::	0		32768	i
*	2001:DB8:ACAD:1001::/64	::	0		32768	i

R1#

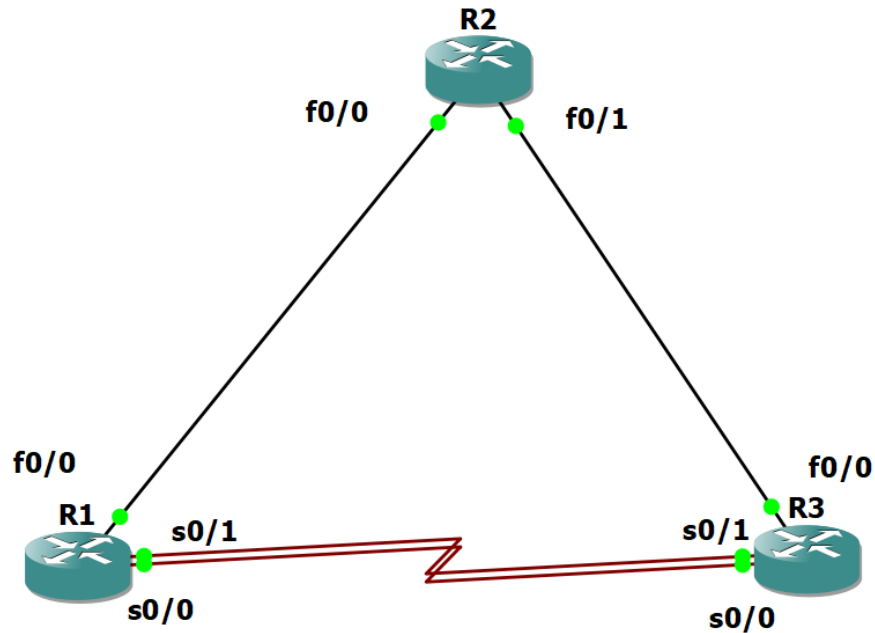
Viewing Router Tables


```
R1#show ipv6 route bgp
IPv6 Routing Table - default - 11 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
        B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
        I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
        EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
        NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
        OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, l - LISP
R1#
```

3. Implement BGP path Manipulation

Part 1: Build the Network and Configure Basic Device Settings and Interface Addressing

Step 1: Topology



Step 2: Configure basic settings for each router. • Router R1

enable

configure terminal

line con 0

exec-timeout 0 0

logging synchronous

exit

banner motd # This is R1, BGP Path Manipulation Lab #

ipv6 unicast-routing

interface FastEthernet0/0

ip address 10.1.2.1 255.255.255.0

ipv6 address FE80::1:1 link-local

ipv6 address 2001:DB8:ACAD:1012::1/64

no shutdown

exit

interface Serial0/0

ip address 10.1.3.1 255.255.255.128

ipv6 address FE80::1:2 link-local

```
ipv6 address 2001:DB8:ACAD:1013::1/64
clock rate 64000
no shutdown
exit
```

```
interface Serial0/1
ip address 10.1.3.129 255.255.255.128
ipv6 address FE80::1:3 link-local
ipv6 address 2001:DB8:ACAD:1014::1/64
clock rate 64000
no shutdown
exit
```

```
interface Loopback0
ip address 192.168.1.1 255.255.255.224
ipv6 address FE80::1:4 link-local
ipv6 address 2001:DB8:ACAD:1000::1/64
no shutdown
exit
```

```
interface Loopback1
ip address 192.168.1.65 255.255.255.192
ipv6 address FE80::1:5 link-local
ipv6 address 2001:DB8:ACAD:1001::1/64
no shutdown
exit
```

```
end
write memory
```

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#line con 0
R1(config-line)# exec-timeout 0 0
R1(config-line)# logging synchronous
R1(config-line)# exit
R1(config)#
R1(config)#banner motd # This is R1, BGP Path Manipulation Lab #
R1(config)#
R1(config)#ipv6 unicast-routing
R1(config)#
R1(config)#interface FastEthernet0/0
R1(config-if)# ip address 10.1.2.1 255.255.255.0
R1(config-if)# ipv6 address FE80::1:1 link-local
R1(config-if)# ipv6 address 2001:DB8:ACAD:1012::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
R1(config)#interface Serial0/0
R1(config-if)# ip address 10.1.3.1 255.255.255.128
R1(config-if)# ipv6 address FE80::1:2 link-local
R1(config-if)# ipv6 address 2001:DB8:ACAD:1013::1/64
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
R1(config)#interface Serial0/1
R1(config-if)# ip address 10.1.3.129 255.255.255.128
R1(config-if)# ipv6 address FE80::1:3 link-local
R1(config-if)# ipv6 address 2001:DB8:ACAD:1014::1/64
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
R1(config)#interface Loopback0
R1(config-if)# ip address 192.168.1.1 255.255.255.224
R1(config-if)# ipv6 address FE80::1:4 link-local
R1(config-if)# ipv6 address 2001:DB8:ACAD:1000::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
R1(config)#interface Loopback1
R1(config-if)# ip address 192.168.1.65 255.255.255.192
R1(config-if)# ipv6 address FE80::1:5 link-local
R1(config-if)# ipv6 address 2001:DB8:ACAD:1001::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
R1(config)#end
```

R2

enable

configure terminal

no ip domain lookup

line con 0

```
exec-timeout 0 0
logging synchronous
exit
```

```
banner motd # This is R2, BGP Path Manipulation Lab #
```

```
ipv6 unicast-routing
```

```
interface FastEthernet0/0
ip address 10.1.2.2 255.255.255.0
ipv6 address FE80::2:1 link-local
ipv6 address 2001:DB8:ACAD:1012::2/64
no shutdown
exit
```

```
interface FastEthernet0/1
ip address 10.2.3.2 255.255.255.0
ipv6 address FE80::2:2 link-local
ipv6 address 2001:DB8:ACAD:1023::2/64
no shutdown
exit
```

```
interface Loopback0
ip address 192.168.2.1 255.255.255.224
ipv6 address FE80::2:3 link-local
ipv6 address 2001:DB8:ACAD:2000::2/64
no shutdown
exit
```

```
interface Loopback1
ip address 192.168.2.65 255.255.255.192
ipv6 address FE80::2:4 link-local
ipv6 address 2001:DB8:ACAD:2001::2/64
no shutdown
exit
```

```
end
write memory
```

```
R2#enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#
R2(config)#no ip domain lookup
R2(config)#
R2(config)#line con 0
R2(config-line)# exec-timeout 0 0
R2(config-line)# logging synchronous
R2(config-line)# exit
R2(config)#
R2(config)#banner motd # This is R2, BGP Path Manipulation Lab #
R2(config)#
R2(config)#ipv6 unicast-routing
R2(config)#
R2(config)#interface FastEthernet0/0
R2(config-if)# ip address 10.1.2.2 255.255.255.0
R2(config-if)# ipv6 address FE80::2:1 link-local
R2(config-if)# ipv6 address 2001:DB8:ACAD:1012::2/64
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)#
R2(config)#interface FastEthernet0/1
R2(config-if)# ip address 10.2.3.2 255.255.255.0
R2(config-if)# ipv6 address FE80::2:2 link-local
R2(config-if)# ipv6 address 2001:DB8:ACAD:1023::2/64
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)#
R2(config)#interface Loopback0
R2(config-if)# ip address 192.168.2.1 255.255.255.224
R2(config-if)# ipv6 address FE80::2:3 link-local
R2(config-if)# ipv6 address 2001:DB8:ACAD:2000::2/64
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)#
R2(config)#interface Loopback1
R2(config-if)# ip address 192.168.2.65 255.255.255.192
R2(config-if)# ipv6 address FE80::2:4 link-local
R2(config-if)# ipv6 address 2001:DB8:ACAD:2001::2/64
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)#
R2(config)#end
```

Router R3

enable

configure terminal

no ip domain lookup

line con 0

exec-timeout 0 0

logging synchronous

exit

banner motd # This is R3, BGP Path Manipulation Lab #

ipv6 unicast-routing

```
interface FastEthernet0/0
ip address 10.2.3.3 255.255.255.0
ipv6 address FE80::3:1 link-local
ipv6 address 2001:DB8:ACAD:1023::3/64
no shutdown
exit
```

```
interface Serial0/0
ip address 10.1.3.3 255.255.255.128
ipv6 address FE80::3:2 link-local
ipv6 address 2001:DB8:ACAD:1013::3/64
clock rate 64000
no shutdown
exit
```

```
interface Serial0/1
ip address 10.1.3.130 255.255.255.128
ipv6 address FE80::3:3 link-local
ipv6 address 2001:DB8:ACAD:1014::3/64
clock rate 64000
no shutdown
exit
```

```
interface Loopback0
ip address 192.168.3.1 255.255.255.224
ipv6 address FE80::3:4 link-local
ipv6 address 2001:DB8:ACAD:3000::1/64
no shutdown
exit
```

```
interface Loopback1
ip address 192.168.3.65 255.255.255.192
ipv6 address FE80::3:5 link-local
ipv6 address 2001:DB8:ACAD:3001::1/64
no shutdown
exit
```

```
end
write memory
```

```

R3#enable
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#
R3(config)#no ip domain lookup
R3(config)#
R3(config)#line con 0
R3(config-line)# exec-timeout 0 0
R3(config-line)# logging synchronous
R3(config-line)# exit
R3(config)#
R3(config)#banner motd # This is R3, BGP Path Manipulation Lab #
R3(config)#
R3(config)#ipv6 unicast-routing
R3(config)#
R3(config)#interface FastEthernet0/0
R3(config-if)# ip address 10.2.3.3 255.255.255.0
R3(config-if)# ipv6 address FE80::3:1 link-local
R3(config-if)# ipv6 address 2001:DB8:ACAD:1023::3/64
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)#
R3(config)#interface Serial0/0
R3(config-if)# ip address 10.1.3.3 255.255.255.128
R3(config-if)# ipv6 address FE80::3:2 link-local
R3(config-if)# ipv6 address 2001:DB8:ACAD:1013::3/64
R3(config-if)# clock rate 64000
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)#
R3(config)#interface Serial0/1
R3(config-if)# ip address 10.1.3.130 255.255.255.128
R3(config-if)# ipv6 address FE80::3:3 link-local
R3(config-if)# ipv6 address 2001:DB8:ACAD:1014::3/64
R3(config-if)# clock rate 64000
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)#
R3(config)#interface Loopback0
R3(config-if)# ip address 192.168.3.1 255.255.255.224
R3(config-if)# ipv6 address FE80::3:4 link-local
R3(config-if)# ipv6 address 2001:DB8:ACAD:3000::1/64
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)#
R3(config)#interface Loopback1
R3(config-if)# ip address 192.168.3.65 255.255.255.192
R3(config-if)# ipv6 address FE80::3:5 link-local
R3(config-if)# ipv6 address 2001:DB8:ACAD:3001::1/64
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)#

```

Part 2: Configure and Verify Multi-Protocol BGP on all Routers

Step 1: On R1, create the core BGP configuration.

```

R1#configure terminal
router bgp 6500
  bgp router-id 1.1.1.1
  no bgp default ipv4-unicast
  neighbor 10.1.2.2 remote-as 500
  neighbor 10.1.3.3 remote-as 300
  neighbor 10.1.3.130 remote-as 300
  neighbor 2001:DB8:ACAD:1012::2 remote-as 500

```



```
neighbor 2001:DB8:ACAD:1013::3 remote-as 300
neighbor 2001:DB8:ACAD:1014::3 remote-as 300
end
write memory
```

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router bgp 6500
R1(config-router)# bgp router-id 1.1.1.1
R1(config-router)# no bgp default ipv4-unicast
R1(config-router)# neighbor 10.1.2.2 remote-as 500
R1(config-router)# neighbor 10.1.3.3 remote-as 300
R1(config-router)# neighbor 10.1.3.130 remote-as 300
R1(config-router)# neighbor 2001:DB8:ACAD:1012::2 remote-as 500
R1(config-router)# neighbor 2001:DB8:ACAD:1013::3 remote-as 300
R1(config-router)# neighbor 2001:DB8:ACAD:1014::3 remote-as 300
R1(config-router)#end
R1#write memory
Building configuration...
```

Step 2: On R1, configure the IPv4 unicast address family.

```
R1#configure terminal
router bgp 6500
address-family ipv4 unicast
network 192.168.1.0 mask 255.255.255.224
network 192.168.1.64 mask 255.255.255.192
no neighbor 2001:DB8:ACAD:1012::2 activate
no neighbor 2001:DB8:ACAD:1013::3 activate
no neighbor 2001:DB8:ACAD:1014::3 activate
neighbor 10.1.2.2 activate
neighbor 10.1.3.3 activate
neighbor 10.1.3.130 activate
exit-address-family
end
write memory
```

```

R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router bgp 6500
R1(config-router)# address-family ipv4 unicast
R1(config-router-af)# network 192.168.1.0 mask 255.255.255.224
R1(config-router-af)# network 192.168.1.64 mask 255.255.255.192
R1(config-router-af)# no neighbor 2001:DB8:ACAD:1012::2 activate
R1(config-router-af)# no neighbor 2001:DB8:ACAD:1013::3 activate
R1(config-router-af)# no neighbor 2001:DB8:ACAD:1014::3 activate
R1(config-router-af)# neighbor 10.1.2.2 activate
R1(config-router-af)# neighbor 10.1.3.3 activate
R1(config-router-af)# neighbor 10.1.3.130 activate
R1(config-router-af)# exit-address-family
R1(config-router)#end
R1#write memory
Building configuration...

```

Step 3: On R1, configure the IPv6 unicast address family.

```

R1#configure terminal
router bgp 6500
address-family ipv6 unicast
network 2001:DB8:ACAD:1000::/64
network 2001:DB8:ACAD:1001::/64
neighbor 2001:DB8:ACAD:1012::2 activate
neighbor 2001:DB8:ACAD:1013::3 activate
neighbor 2001:DB8:ACAD:1014::3 activate
exit-address-family
end
write memory

```

```

R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router bgp 6500
R1(config-router)# address-family ipv6 unicast
R1(config-router-af)# network 2001:DB8:ACAD:1000::/64
R1(config-router-af)# network 2001:DB8:ACAD:1001::/64
R1(config-router-af)# neighbor 2001:DB8:ACAD:1012::2 activate
R1(config-router-af)# neighbor 2001:DB8:ACAD:1013::3 activate
R1(config-router-af)# neighbor 2001:DB8:ACAD:1014::3 activate
R1(config-router-af)# exit-address-family
R1(config-router)#end
R1#write memory
Building configuration...

```

Step 4: Configure MP-BGP on R2 and R3 as you did in the previous step.

```

conf t
router bgp 500
bgp router-id 2.2.2.2
no bgp default ipv4-unicast

```

```
neighbor 10.1.2.1 remote-as 6500
neighbor 10.2.3.3 remote-as 300
neighbor 2001:db8:acad:1012::1 remote-as 6500
neighbor 2001:db8:acad:1023::3 remote-as 300
```

```
address-family ipv4
network 192.168.2.0 mask 255.255.255.224
network 192.168.2.64 mask 255.255.255.192
neighbor 10.1.2.1 activate
neighbor 10.2.3.3 activate
exit-address-family
```

```
address-family ipv6
network 2001:db8:acad:2000::/64
network 2001:db8:acad:2001::/64
neighbor 2001:db8:acad:1012::1 activate
neighbor 2001:db8:acad:1023::3 activate
exit-address-family
```

end

Wr

```
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router bgp 500
R2(config-router)# bgp router-id 2.2.2.2
R2(config-router)# no bgp default ipv4-unicast
R2(config-router)#
R2(config-router)# neighbor 10.1.2.1 remote-as 6500
R2(config-router)# neighbor 10.2.3.3 remote-as 300
R2(config-router)# neighbor 2001:db8:acad:1012::1 remote-as 6500
R2(config-router)# neighbor 2001:db8:acad:1023::3 remote-as 300
R2(config-router)#
R2(config-router)# address-family ipv4
R2(config-router-af)# network 192.168.2.0 mask 255.255.255.224
R2(config-router-af)# network 192.168.2.64 mask 255.255.255.192
R2(config-router-af)# neighbor 10.1.2.1 activate
R2(config-router-af)# neighbor 10.2.3.3 activate
R2(config-router-af)# exit-address-family
R2(config-router)#
R2(config-router)# address-family ipv6
R2(config-router-af)# network 2001:db8:acad:2000::/64
R2(config-router-af)# network 2001:db8:acad:2001::/64
R2(config-router-af)# neighbor 2001:db8:acad:1012::1 activate
R2(config-router-af)# neighbor 2001:db8:acad:1023::3 activate
R2(config-router-af)# exit-address-family
R2(config-router)#e
*Mar 1 00:18:12.563: %BGP-5-ADJCHANGE: neighbor 10.1.2.1 Up
R2(config-router)#end
```

Router R3:

```
conf t
router bgp 300
bgp router-id 3.3.3.3
no bgp default ipv4-unicast

neighbor 10.1.3.1 remote-as 6500
neighbor 10.1.3.129 remote-as 6500
```

```
neighbor 10.2.3.2 remote-as 500
neighbor 2001:db8:acad:1013::1 remote-as 6500
neighbor 2001:db8:acad:1014::1 remote-as 6500
neighbor 2001:db8:acad:1023::2 remote-as 500
```

```
address-family ipv4
network 192.168.3.0 mask 255.255.255.224
network 192.168.3.64 mask 255.255.255.192
neighbor 10.1.3.1 activate
neighbor 10.1.3.129 activate
neighbor 10.2.3.2 activate
exit-address-family
```

```
address-family ipv6
network 2001:db8:acad:3000::/64
network 2001:db8:acad:3001::/64
neighbor 2001:db8:acad:1013::1 activate
neighbor 2001:db8:acad:1014::1 activate
neighbor 2001:db8:acad:1023::2 activate
exit-address-family
```

end

Wr

```
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router bgp 300
R3(config-router)# bgp router-id 3.3.3.3
R3(config-router)# no bgp default ipv4-unicast
R3(config-router)#
R3(config-router)# neighbor 10.1.3.1 remote-as 6500
R3(config-router)# neighbor 10.1.3.129 remote-as 6500
R3(config-router)# neighbor 10.2.3.2 remote-as 500
R3(config-router)# neighbor 2001:db8:acad:1013::1 remote-as 6500
R3(config-router)# neighbor 2001:db8:acad:1014::1 remote-as 6500
R3(config-router)# neighbor 2001:db8:acad:1023::2 remote-as 500
R3(config-router)#
R3(config-router)# address-family ipv4
R3(config-router-af)# network 192.168.3.0 mask 255.255.255.224
R3(config-router-af)# network 192.168.3.64 mask 255.255.255.192
R3(config-router-af)# neighbor 10.1.3.1 activate
R3(config-router-af)# neighbor 10.1.3.129 activate
R3(config-router-af)# neighbor 10.2.3.2 activate
R3(config-router-af)# exit-address-family
R3(config-router)#
R3(config-router)# address-family ipv6
R3(config-router-af)# network 2001:db8:acad:3000::/64
R3(config-router-af)# network 2001:db8:acad:3001::/64
R3(config-router-af)# neighbor 2001:db8:acad:1013::1 activate
R3(config-router-af)# neighbor 2001:db8:acad:1014::1 activate
R3(config-router-af)# neighbor 2001:db8:acad:1023::2 activate
R3(config-router-af)# exit-address-family
R3(config-router)#
R3(config-router)#end
```

Step 5: Verify that MP-BGP is operational.

R1#show bgp ipv4 unicast summary

```
R1#show bgp ipv4 unicast summary
BGP router identifier 1.1.1.1, local AS number 6500
BGP table version is 7, main routing table version 7
6 network entries using 720 bytes of memory
14 path entries using 728 bytes of memory
6/3 BGP path/bestpath attribute entries using 744 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 2320 total bytes of memory
BGP activity 12/0 prefixes, 28/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
10.1.2.2       4   500      9      9       7   0   0 00:03:10      4
10.1.3.3       4   300      8      8       7   0   0 00:01:58      4
10.1.3.130    4   300      8      8       7   0   0 00:01:51      4
```

R1#show bgp ipv6 unicast summary

```
R1#show bgp ipv6 unicast summary
BGP router identifier 1.1.1.1, local AS number 6500
BGP table version is 7, main routing table version 7
6 network entries using 912 bytes of memory
14 path entries using 1064 bytes of memory
6/3 BGP path/bestpath attribute entries using 744 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 2848 total bytes of memory
BGP activity 12/0 prefixes, 28/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
2001:DB8:ACAD:1012::2
  4   500      10     10       7   0   0 00:03:52      4
2001:DB8:ACAD:1013::3
  4   300       9      9       7   0   0 00:02:59      4
2001:DB8:ACAD:1014::3
  4   300       9      9       7   0   0 00:02:55      4
% NOTE: This command is deprecated. Please use 'show bgp ipv6 unicast'
```

R1#show bgp ipv4 unicast | begin Network

```
R1#show bgp ipv4 unicast | begin Network
Network      Next Hop      Metric LocPrf Weight Path
*> 192.168.1.0/27  0.0.0.0          0         32768 i
*> 192.168.1.64/26 0.0.0.0          0         32768 i
* 192.168.2.0/27  10.1.3.130      0         0 300 500 i
*                  10.1.3.3         0         0 300 500 i
*>                  10.1.2.2         0         0 500 i
* 192.168.2.64/26 10.1.3.130      0         0 300 500 i
*                  10.1.3.3         0         0 300 500 i
*>                  10.1.2.2         0         0 500 i
* 192.168.3.0/27  10.1.2.2         0         0 500 300 i
*                  10.1.3.130      0         0 300 i
*>                  10.1.3.3         0         0 300 i
* 192.168.3.64/26 10.1.2.2         0         0 500 300 i
*                  10.1.3.130      0         0 300 i
*>                  10.1.3.3         0         0 300 i
R1#
```

R1#show bgp ipv6 unicast | begin Network

```

R1#show bgp ipv6 unicast | begin Network
  Network      Next Hop      Metric LocPrf Weight Path
*> 2001:DB8:ACAD:1000::/64
      ::              0          32768 i
*> 2001:DB8:ACAD:1001::/64
      ::              0          32768 i
* 2001:DB8:ACAD:2000::/64
      2001:DB8:ACAD:1014::3
                                0 300 500 i
*
      2001:DB8:ACAD:1013::3
                                0 300 500 i
*>
      2001:DB8:ACAD:1012::2
                                0 500 i
* 2001:DB8:ACAD:2001::/64
      2001:DB8:ACAD:1014::3
                                0 300 500 i
*
      2001:DB8:ACAD:1013::3
                                0 300 500 i
*>
      2001:DB8:ACAD:1012::2
                                0 500 i
* 2001:DB8:ACAD:3000::/64
      2001:DB8:ACAD:1012::2
                                0 500 300 i
*
      2001:DB8:ACAD:1014::3
                                0 300 i
*>
      2001:DB8:ACAD:1013::3
                                0 300 i
* 2001:DB8:ACAD:3001::/64
      2001:DB8:ACAD:1012::2
                                0 500 300 i
*
      2001:DB8:ACAD:1014::3
                                0 300 i
*>
      2001:DB8:ACAD:1013::3
                                0 300 i

```

• View Routing Tables:

R1#show ipv6 route bgp

```

R1#show ipv6 route bgp
IPv6 Routing Table - 15 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
B 2001:DB8:ACAD:2000::/64 [20/0]
  via FE80::2:1, FastEthernet0/0
B 2001:DB8:ACAD:2001::/64 [20/0]
  via FE80::2:1, FastEthernet0/0
B 2001:DB8:ACAD:3000::/64 [20/0]
  via FE80::3:2, Serial0/0
B 2001:DB8:ACAD:3001::/64 [20/0]
  via FE80::3:2, Serial0/0

```

Part 3: Configure and Verify BGP Path Manipulation Settings on all Routers

Step 1: Configure ACL-based route filtering.

On R1, issue the command `show bgp ipv4 unicast i 300` to see what prefixes ASN300 sharing via BGP. Take note of those prefixes that do not originate in ASN300.

R1#show bgp ipv4 unicast | i 300

```
R1#show bgp ipv4 unicast | i 300
* 192.168.2.0/27 10.1.3.130 0 300 500 i
* 10.1.3.3 0 300 500 i
* 192.168.2.64/26 10.1.3.130 0 300 500 i
* 10.1.3.3 0 300 500 i
* 192.168.3.0/27 10.1.2.2 0 500 300 i
* 10.1.3.130 0 0 300 i
*> 10.1.3.3 0 0 300 i
* 192.168.3.64/26 10.1.2.2 0 500 300 i
* 10.1.3.130 0 0 300 i
*> 10.1.3.3 0 0 300 i
```

R1#show bgp ipv4 unicast begin 192.168.3

```
R1#show bgp ipv4 unicast | begin 192.168.3
* 192.168.3.0/27 10.1.2.2 0 500 300 i
* 10.1.3.130 0 0 300 i
*> 10.1.3.3 0 0 300 i
* 192.168.3.64/26 10.1.2.2 0 500 300 i
* 10.1.3.130 0 0 300 i
*> 10.1.3.3 0 0 300 i
R1#
```

- On R3, configure an access list designed to match the source address and mask of the networks belonging to ASN300:

```
conf t
ip access-list extended ALLOWED_TO_R!
permit ip 192.168.3.0 0.0.0.0 255.255.255.224 0.0.0.0
permit ip 192.168.3.64 0.0.0.0 255.255.255.192 0.0.0.0
exit
end
Wr
```

```
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip access-list extended ALLOWED_TO_R!
R3(config-ext-nacl)# permit ip 192.168.3.0 0.0.0.0 255.255.255.224 0.0.0.0
R3(config-ext-nacl)# permit ip 192.168.3.64 0.0.0.0 255.255.255.192 0.0.0.0
R3(config-ext-nacl)#exit
R3(config)#end
```

- On R3, apply the ALLOWED_TO_R1 ACL as a distribute list to the IPv4 neighbor adjacencies with R1.

```
enable
configure terminal

router bgp 300
address-family ipv4 unicast
neighbor 10.1.3.1 distribute-list ALLOWED_TO_R1 out
neighbor 10.1.3.129 distribute-list ALLOWED_TO_R1 out
exit-address-family

end
write memory
```



```

R3#enable
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#
R3(config)#router bgp 300
R3(config-router)# address-family ipv4 unicast
R3(config-router-af)# neighbor 10.1.3.1 distribute-list ALLOWED_TO_R1 out
R3(config-router-af)# neighbor 10.1.3.129 distribute-list ALLOWED_TO_R1 out
R3(config-router-af)# exit-address-family
R3(config-router)#
R3(config-router)#end
R3#write memory
Building configuration...

```

Perform a reset of the IPv4 adjacency with R1 for the outbound traffic without tearing down the session.

```
R3#clear bgp ipv4 unicast 6500 out
```

```

R3#clear bgp ipv4 unicast 6500 out
R3#

```

On R1, issue the command `show bgp ipv4 unicast i 300` to see what prefixes routes ASN300

is now sharing via BGP. All of the prefixes should now originate in ASN300:

```
R1#show bgp ipv4 unicast | i 300
```

```

R1#show bgp ipv4 unicast | i 300
* 192.168.2.0/27 10.1.3.130 0 300 500 i
* 10.1.3.3 0 300 500 i
* 192.168.2.64/26 10.1.3.130 0 300 500 i
* 10.1.3.3 0 300 500 i
* 192.168.3.0/27 10.1.2.2 0 500 300 i
* 10.1.3.130 0 0 300 i
*> 10.1.3.3 0 0 300 i
* 192.168.3.64/26 10.1.2.2 0 500 300 i
* 10.1.3.130 0 0 300 i
*> 10.1.3.3 0 0 300 i

```

Step 2: Configure prefix-list-based route filtering.

On R1, issue the command `show bgp ipv4 unicast begin 192.168.3` to see what prefixes ASN500 is sharing via BGP. Take note of those prefixes that do not originate in ASN500.

```
R1#show bgp ipv4 unicast | begin 192.168.3
```

```

R1#show bgp ipv4 unicast | begin 192.168.3
* 192.168.3.0/27 10.1.2.2 0 500 300 i
* 10.1.3.130 0 0 300 i
*> 10.1.3.3 0 0 300 i
* 192.168.3.64/26 10.1.2.2 0 500 300 i
* 10.1.3.130 0 0 300 i
*> 10.1.3.3 0 0 300 i

```

On R1, configure AS-PATH ACL to match the routes from the local ASN.


```
ip prefix-list ALLOWED_FROM_R2 seq 5 permit 192.168.2.0/24 le 27
```

```
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip prefix-list ALLOWED_FROM_R2 seq 5 permit 192.168.2.0/24 le 27
R1(config)#
```

On R1, apply the AS-PATH ACL as a filter-list on the adjacency configured with R2.

```
conf t
router bgp 6500
address-family ipv4 unicast
neighbor 10.1.2.2 prefix-list ALLOWED_FROM_R2 in
exit-address-family
end
wr
```

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router bgp 6500
R1(config-router)# address-family ipv4 unicast
R1(config-router-af)# neighbor 10.1.2.2 prefix-list ALLOWED_FROM_R2 in
R1(config-router-af)#
*Mar  1 00:39:45.231: %SYS-5-CONFIG_I: Configured from console by console
R1(config-router-af)#
```

On R1, perform a reset of the IPv4 adjacency with R2 for the outbound traffic without tearing down the session

```
R1#clear bgp ipv4 unicast 500 in
```

```
R1#clear bgp ipv4 unicast 500 in
R1#
```

• On R1, issue the command `show bgp ipv4 unicast i 500` to see what prefixes routes ASN500 is now sharing via BGP. All of the prefixes should now originate in ASN500.

```
R1#show bgp ipv4 unicast | i 500
```

```
R1#show bgp ipv4 unicast | i 500
* 192.168.2.0/27 10.1.3.130 0 300 500 i
* 10.1.3.3 0 300 500 i
*> 10.1.2.2 0 0 500 i
* 192.168.2.64/26 10.1.3.130 0 300 500 i
* 10.1.3.3 0 300 500 i
*> 10.1.2.2 0 0 500 i
```

• On R2, issue the command `show bgp ipv4 unicast i 6500` to see what prefixes routes ASN6500 is now sharing via BGP. All of the prefixes should now originate in ASN6500.

Step 3: Configure an AS-PATH ACL to filter routes being advertised.

• On R2, issue the command `show bgp ipv4 unicast begin Network` to see what prefixes ASN6500 is sharing via BGP.

R2#show bgp ipv4 unicast | begin Network

```
R2#show bgp ipv4 unicast | begin Network
  Network          Next Hop           Metric LocPrf Weight Path
* 192.168.1.0/27    10.2.3.3                          0 300 6500 i
*>                  10.1.2.1                          0      6500 i
* 192.168.1.64/26   10.2.3.3                          0 300 6500 i
*>                  10.1.2.1                          0      6500 i
*> 192.168.2.0/27    0.0.0.0                          0    32768 i
*> 192.168.2.64/26   0.0.0.0                          0    32768 i
* 192.168.3.0/27    10.1.2.1                          0 6500 300 i
*>                  10.2.3.3                          0      300 i
* 192.168.3.64/26   10.1.2.1                          0 6500 300 i
*>                  10.2.3.3                          0      300 i
```

- On R1, configure AS-PATH ACL to match the routes from the local ASN.

conf t

ip as-path access-list 1 permit ^\$

end

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# ip as-path access-list 1 permit ^$
R1(config)#end
R1#
```

- On R1, apply the AS-PATH ACL as a filter-list on the adjacency configured with R2.

conf t

router bgp 6500

address-family ipv4 unicast

neighbor 10.1.2.2 filter-list 1 out

end

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router bgp 6500
R1(config-router)# address-family ipv4 unicast
R1(config-router-af)# neighbor 10.1.2.2 filter-list 1 out
R1(config-router-af)#end
R1#
```

- On R1, perform a reset of the IPv4 adjacency with R2 for the outbound traffic without tearing down the session

R1#clear bgp ipv4 unicast 500 out

```
R1#clear bgp ipv4 unicast 500 out
R1#
```

- On R2, issue the command show bgp ipv4 unicast i 6500 to see what prefixes routes ASN6500 is now sharing via BGP. All of the prefixes should now originate in ASN6500.

R2#show bgp ipv4 unicast i 6500

```
R2#show bgp ipv4 unicast | i 6500
* 192.168.1.0/27 10.2.3.3 0 300 6500 i
*> 10.1.2.1 0 6500 i
* 192.168.1.64/26 10.2.3.3 0 300 6500 i
*> 10.1.2.1 0 6500 i
R2#
```

Step 4: Configure IPv6 prefix-list-based route filtering.

- On R1, issue the command `show bgp ipv6 unicast neighbors 2001:db8:acad:1012::2 routes` to see what IPv6 prefixes ASN500 is sharing via BGP.

show bgp ipv6 unicast neighbors 2001:db8:acad:1012::2 routes

```
R1#show bgp ipv6 unicast neighbors 2001:db8:acad:1012::2 routes
BGP table version is 7, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop          Metric LocPrf Weight Path
*> 2001:DB8:ACAD:2000::/64
                        2001:DB8:ACAD:1012::2
                                                0          0 500 i
*> 2001:DB8:ACAD:2001::/64
                        2001:DB8:ACAD:1012::2
                                                0          0 500 i
* 2001:DB8:ACAD:3000::/64
                        2001:DB8:ACAD:1012::2
                                                0 500 300 i
* 2001:DB8:ACAD:3001::/64
                        2001:DB8:ACAD:1012::2
                                                0 500 300 i

Total number of prefixes 4
```

- On R1, configure an IPv6 prefix list designed to match the source address and mask of networks belonging to ASN500.

conf t

```
ipv6 prefix-list IPV6_ALLOWED_FROM_R2 seq 5 permit 2001:db8:acad:2000::/64
ipv6 prefix-list IPV6_ALLOWED_FROM_R2 seq 10 permit 2001:db8:acad:2001::/64
end
```

wr

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#$-list IPV6_ALLOWED_FROM_R2 seq 5 permit 2001:db8:acad:2000::/64
R1(config)#$-list IPV6_ALLOWED_FROM_R2 seq 10 permit 2001:db8:acad:2001::/64
R1(config)#end
R1#wr
Building configuration...
```

Apply the IPV6_ALLOWED_FROM_R2 prefix list to the IPv6 neighbor adjacencies for R2

conf t

router bgp 6500

address-family ipv6 unicast

neighbor 2001:db8:acad:1012::2 prefix-list IPV6_ALLOWED_FROM_R2 in

exit-address-family

end

wr

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router bgp 6500
R1(config-router)# address-family ipv6 unicast
R1(config-router-af)# $cad:1012::2 prefix-list IPV6_ALLOWED_FROM_R2 in
R1(config-router-af)# exit-address-family
R1(config-router)#end
R1#wr
Building configuration...
```

- Perform a reset of the IPv6 adjacency with R2 for the inbound traffic without tearing down the session.

R1#clear bgp ipv6 unicast 500 in

```
R1#clear bgp ipv6 unicast 500 in
R1#
```

- On R1, issue the command show bgp ipv6 unicast neighbors 2001:db8:acad:1012::2 routes to see what IPv6 prefixes routes ASN500 is now sharing via BGP. All of the IPv6 prefixes should now originate in ASN500.

R1#show bgp ipv6 unicast neighbors 2001: db8: acad:1012::2 routes

```
R1#show bgp ipv6 unicast neighbors 2001:db8:acad:1012::2 routes
BGP table version is 7, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network                Next Hop                Metric LocPrf Weight Path
*> 2001:DB8:ACAD:2000::/64
                        2001:DB8:ACAD:1012::2
                                0                      0 500 i
*> 2001:DB8:ACAD:2001::/64
                        2001:DB8:ACAD:1012::2
                                0                      0 500 i

Total number of prefixes 2
```

Step 5: Configure BGP path attribute manipulation to effect routing

R1#show bgp ipv4 unicast 192.168.3.0

```
R1#show bgp ipv4 unicast 192.168.3.0
BGP routing table entry for 192.168.3.0/27, version 7
Paths: (2 available, best #2, table Default-IP-Routing-Table)
  Not advertised to any peer
  300
    10.1.3.130 from 10.1.3.130 (3.3.3.3)
      Origin IGP, metric 0, localpref 100, valid, external
  300
    10.1.3.3 from 10.1.3.3 (3.3.3.3)
      Origin IGP, metric 0, localpref 100, valid, external, best
```

- On R1, configure a prefix list designed to match the source address and mask of networks belonging to ASN300.

conf t

ip prefix-list PREFERRED_IPV4_PATH seq 5 permit 192.168.3.0/24 le 27

end

WR

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#$ist PREFERRED_IPV4_PATH seq 5 permit 192.168.3.0/24 le 27
R1(config)#end
R1#wr
Building configuration...
```

- Create a route-map named **USE_THIS_PATH_FOR_IPV4** that matches on the prefix list you just created and sets the local preference to 250.

```
conf t
route-map USE_THIS_PATH_FOR_IPV4 permit 10
match ip address prefix-list PREFERRED_IPV4_PATH
set local-preference 250
exit
end
wr
```

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#route-map USE_THIS_PATH_FOR_IPV4 permit 10
R1(config-route-map)# match ip address prefix-list PREFERRED_IPV4_PATH
R1(config-route-map)# set local-preference 250
R1(config-route-map)#exit
R1(config)#end
R1#wr
Building configuration...
[OK]
```

- Next, apply this route map to the BGP neighbor 10.1.3.130. **LITLET Comiguration Commands, Une per e**

```
conf t
router bgp 6500
address-family ipv4 unicast
neighbor 10.1.3.130 route-map USE_THIS_PATH_FOR_IPV4 in
exit-address-family
end
wr
```

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router bgp 6500
R1(config-router)# address-family ipv4 unicast
R1(config-router-af)#$10.1.3.130 route-map USE_THIS_PATH_FOR_IPV4 in
R1(config-router-af)# exit-address-family
R1(config-router)#end
R1#wr
Building configuration...
[OK]
R1#
```

- Perform a reset of the IPv4 adjacency with R3 for the inbound traffic without tearing down the session.

R1#clear bgp ipv4 unicast 300 in

```
R1#clear bgp ipv4 unicast 300 in
R1#
```

- On R1, issue the command `show ip route bgp` and take note of the next hop addresses for the 192.168.3.0/27 and 192.168.3.64/26 networks; it should be 10.1.3.130 for both. Issue the command `show bgp ipv4 unicast` and you should see the local preference value in the appropriate column.

R1#show ip route bgp

```
R1#show ip route bgp
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
B       192.168.2.64/26 [20/0] via 10.1.2.2, 00:37:51
B       192.168.2.0/27 [20/0] via 10.1.2.2, 00:37:51
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
B       192.168.3.64/26 [20/0] via 10.1.3.130, 00:00:03
B       192.168.3.0/27 [20/0] via 10.1.3.130, 00:00:03
```

R1#show bgp ipv4 unicast begin Network

```
R1#show bgp ipv4 unicast | begin Network
      Network          Next Hop           Metric LocPrf Weight Path
*> 192.168.1.0/27      0.0.0.0              0         32768 i
*> 192.168.1.64/26    0.0.0.0              0         32768 i
* 192.168.2.0/27      10.1.3.3              0         0 300 500 i
*>                    10.1.2.2              0         0 500 i
* 192.168.2.64/26     10.1.3.3              0         0 300 500 i
*>                    10.1.2.2              0         0 500 i
*> 192.168.3.0/27      10.1.3.130            0        250    0 300 i
*                    10.1.3.3              0         0 300 i
*> 192.168.3.64/26     10.1.3.130            0        250    0 300 i
*                    10.1.3.3              0         0 300 i
```

• Final Config R1:

R1#show run

```
R1#show run
Building configuration...

Current configuration : 3481 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R1
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
memory-size iomem 5
no ip icmp rate-limit unreachable
ip cef
!
!
!
no ip domain lookup
ipv6 unicast-routing
!
multilink bundle-name authenticated
!
```

```

!
banner motd ^C This is R1, BGP Path Manipulation Lab ^C
!
line con 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line aux 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line vty 0 4
  login
!

```

Final Config R2:

Building configuration...

R2#show run

```

R2#show run
Building configuration...

Current configuration : 2501 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R2
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
memory-size iomem 5
no ip icmp rate-limit unreachable
ip cef
!
!
!
no ip domain lookup
ipv6 unicast-routing
!
multilink bundle-name authenticated
!
banner motd ^C This is R2, BGP Path Manipulation Lab ^C
!
line con 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line aux 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line vty 0 4
  login
!

```

• Final Config R3:

R3#show run

```
R3(config-router)#end
R3#write memory
Building configuration...

*Mar  1 00:29:28.439: %SYS-5-CONFIG_I: Configured from console by console[OK]
R3#clear bgp ipv4 unicast 6500 out
R3#show run
Building configuration...

Current configuration : 2992 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R3
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
memory-size iomem 5
no ip icmp rate-limit unreachable
ip cef
!
!
!
no ip domain lookup
ipv6 unicast-routing
!
multilink bundle-name authenticated

banner motd ^C This is R3, BGP Path Manipulation Lab ^C
!
line con 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line aux 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line vty 0 4
  login
!
!
end
```